

THE FURST OPTICAL PERFORMANCE SECTION

ROY T. SMART

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ABSTRACT

This document is not the FURST instrument paper. It is a preview of the one section of it which is generated from the instrument model, built so that the section can be read and checked on its own before it is delivered to the manuscript. The numbered subsections above it are empty placeholders, standing in for the sections of the manuscript which precede it.

1. INSTRUMENT OVERVIEW

1.1. *Design Rationale*

1.2. *Optical Layout*

1.3. *Optical Performance*

The performance of the optical design was evaluated by tracing rays through an idealized model of the instrument, `furst-optics` (Smart 2026b), built with the open-source `optika` raytracing package (Smart and Kankelborg 2026). The model is shown in Figure 1 and its parameters are listed in Table 1. It places the seven feed optics, the grating, and the detector at their design positions on the Rowland circle. The grating has a radius of curvature of 1354 mm and a ruling density of 2200 mm⁻¹, so the Rowland circle has a radius of 677 mm. The Sun is the field stop of the system and the grating is its pupil stop: the field of view is the solar disk at its largest apparent size, and every ray leaving a point on the disk is directed at a point on the clear aperture of the grating, which is 180 mm wide and 28.3 mm tall. Each feed optic is a convex cylinder of radius 3 mm whose clear aperture subtends 45° of the cylinder and is 20.8 mm tall.

The feed optics illuminate the grating at angles of incidence from 10.46° to 17.93°, and the detector lies 4.99° from the grating normal on the same side, so that the shortest wavelengths are observed close to the Littrow configuration. Each channel therefore places a band about 10.3 nm wide on the detector, and the seven bands together cover 116.9 nm to 184.5 nm, with adjacent channels overlapping by at least 0.6 nm so that the spectrum can be stitched together. The dispersion at the detector

is 5.02 pm per pixel.

Since the virtual image of the entire Sun formed by a feed optic is narrower than a pixel, the quantity that governs the spectral resolution is the disk-integrated line spread function (LSF): the distribution along the dispersion direction of the light from the whole solar disk and the whole pupil at a single wavelength. We compute it by tracing 5 wavelengths spanning the detector in each channel, with the solar disk sampled by a stratified random grid of 15 × 15 points and the pupil by one of 15 × 15 points. The result is shown in Figure 2: the image of the Sun at each wavelength is a line about a pixel wide, taller than the detector, and its profile along the dispersion direction is roughly Gaussian. We take the width of the LSF to be the standard deviation along the dispersion direction of the positions of the rays reaching the detector, added in quadrature with the standard deviation of a uniform distribution one pixel wide, to account for the finite size of the pixels. This width is shown in the top panel of Figure 3 and ranges from 0.40 to 0.72 pixels. It grows toward the long-wavelength channels, whose feed optics sit farthest from the axis of the instrument, as the aberrations of the spherical grating grow with the angle of incidence.

The resolving power, taken as the wavelength divided by twice the width of the LSF, is shown in the bottom panel of Figure 3. It ranges from 24,800 to 37,900 over the traced wavelengths, with a mean of 30,800, and so exceeds the requirement of 2 × 10⁴ throughout the bandpass.

Every figure and numerical value in this section is computed from the instrument model by `furst-instrument-paper` (Smart 2026a), so that they cannot drift from it.

REFERENCES

Roy T. Smart. `furst-instrument-paper`, 2026a. URL <https://github.com/roytsmart/furst-instrument-paper>.

Roy T. Smart. `furst-optics`, 2026b. URL <https://furst-optics.readthedocs.io>.

Roy T. Smart and Charles C. Kankelborg. `Optika`, 2026. URL <https://optika.readthedocs.io>.

TABLE 1

OPTICAL DESIGN PARAMETERS OF THE INSTRUMENT MODEL USED IN THIS SECTION, BESIDE THE VALUES IN TABLE 3 OF THE DRAFT OF SEPTEMBER 18, 2026. ENTRIES IN BOLD DIFFER BETWEEN THE TWO AND NEED TO BE RECONCILED BEFORE SUBMISSION.

Parameter	Draft	Model
Wavelength range	116 to 185 nm	116.9 nm to 184.5 nm
Resolving power	> 20,000	24,800 to 37,900
Number of feed optics	7	7
Radius of feed optic	3 mm	3 mm
Height of feed optic	38 mm	20.8 mm (clear aperture)
Feed optic positions (from grating normal)	10.5° to 17.9°	10.46° to 17.93°
Radius of grating	1359 mm	1354 mm
UV section of grating	200 mm × 30 mm	180 mm × 28.3 mm (clear aperture)
Groove density	1.97 gr/μm	2200 mm ⁻¹
Detector type	CCD230-42	CCD230-42
Detector format	2048 × 1024	2048 × 1032
Pixel size	15 μm	15 μm
CCD position (from grating normal)	4.98°	4.99°

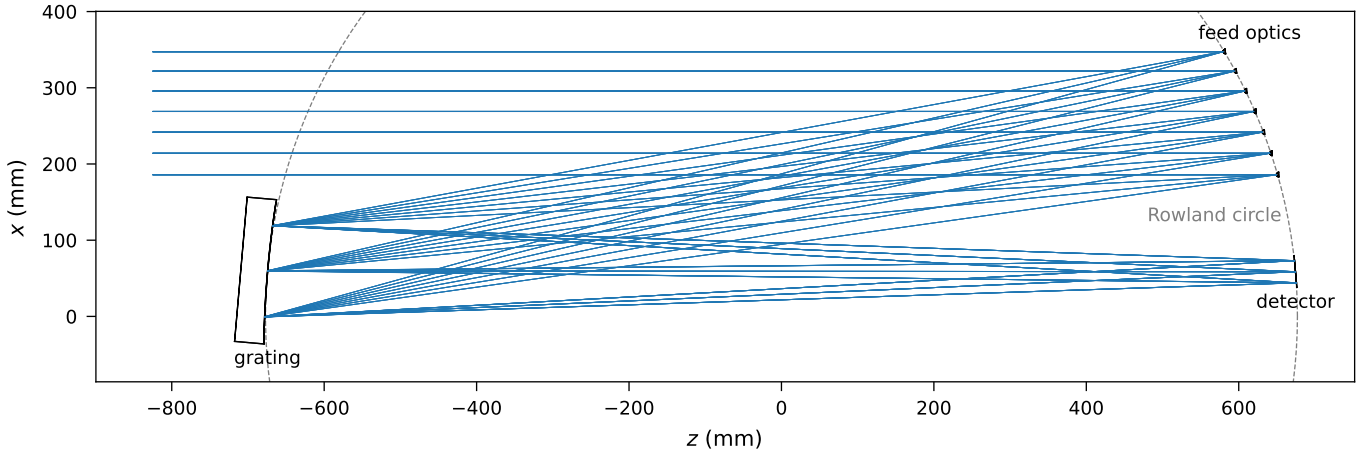


FIG. 1.— The optical layout of FURST, seen from above the optical table. Sunlight enters from the left and is reflected by one of seven convex cylindrical feed optics, each of which forms a virtual image of the Sun on the Rowland circle (dashed), narrower than a pixel of the detector. The concave diffraction grating, at the far side of the circle, disperses the light and focuses it back onto the circle at the detector. Each feed optic illuminates the grating at a different angle of incidence, so each channel places a different band of the spectrum on the detector. Three wavelengths are traced through each channel.

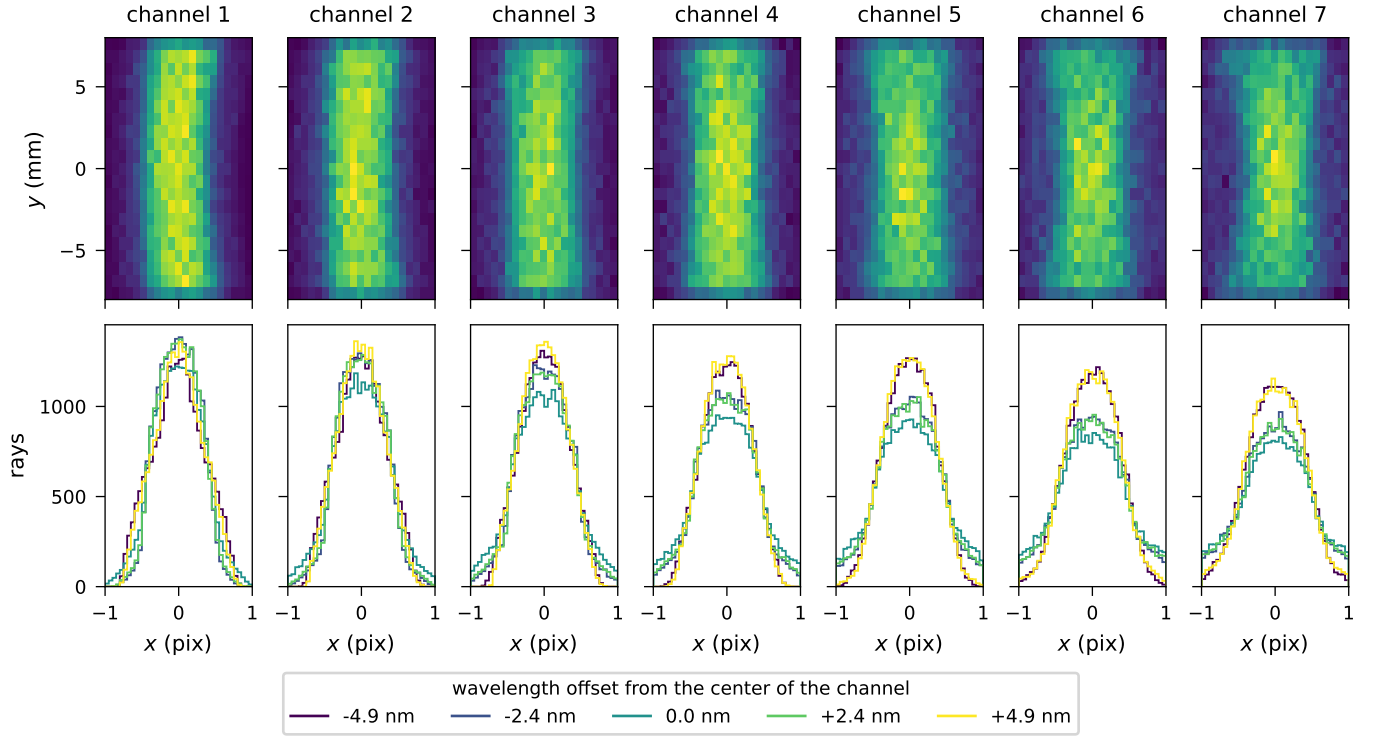


FIG. 2.— The disk-integrated line spread function of each channel of FURST, from a raytrace of the instrument model. Top: the distribution of the rays on the detector at the central traced wavelength of each channel, with the dispersion direction, x , measured in pixels from the mean position of that wavelength and the height on the detector, y , measured from its center. Bottom: the same distribution summed along the height of the detector, at every traced wavelength of each channel, from the shortest (dark) to the longest (light).

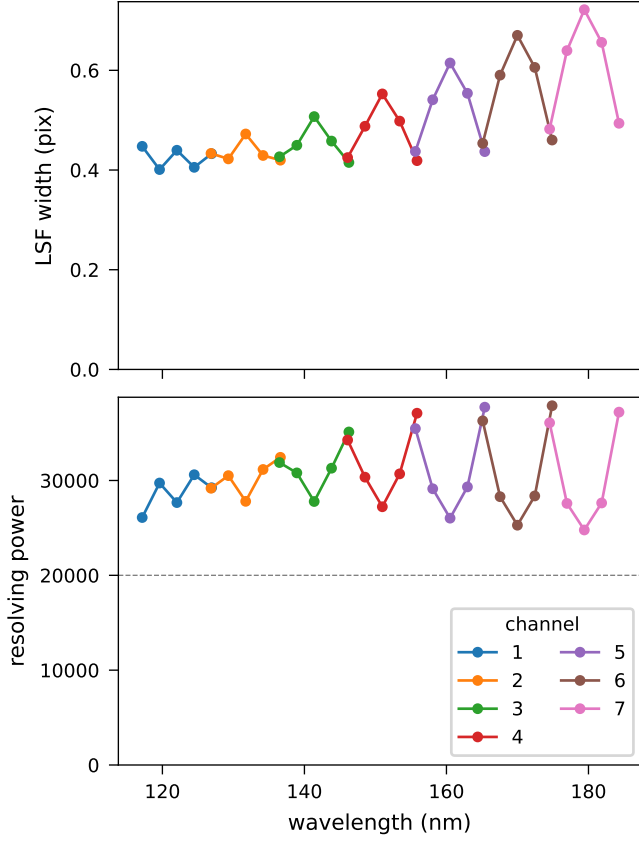


FIG. 3.— The spectral resolution of each channel of FURST at the traced wavelengths, computed from a raytrace of the instrument model. Top: the width of the disk-integrated line spread function, the standard deviation of the positions of the rays along the dispersion direction, added in quadrature with the standard deviation of a uniform distribution one pixel wide. Bottom: the resolving power, taking the resolvable wavelength difference to be twice that width. The dashed line marks the requirement of 2×10^4 .